

PROMPT TEMPLATE LIBRARY

AI Smart Tutor using Advanced Prompt Engineering Techniques

Rancang Bangun Website AI Smart Tutor sebagai Media Tanya Jawab Pembelajaran Interaktif

Field	Information
Prepared by	Ivansyah Putra
Institution / Campus	STMIK Kaputama Binjai
Class / Course	Google Prompt Engineering
Project Type	Web-Based AI Educational Chatbot
Main Technology	Python Flask, Gemini API, HTML, CSS, JavaScript
Data Approach	No Database; Chat History stored using Browser Local Storage
Version	Final Prompt Template Library

This document contains the prompt templates, prompt variables, few-shot examples, and optimization notes used in the AI Smart Tutor website.

Institution / Campus: STMIK Kaputama Binjai

1. Purpose

This Prompt Template Library documents the prompt engineering structure used in the AI Smart Tutor website. The project was prepared by Ivansyah Putra from STMIK Kaputama Binjai for the Google Prompt Engineering class/course. Each AI Tutor Mode uses a different prompt template to ensure structured, relevant, and personalized responses.

- M1: project setup and initial prompt exploration.
- M2: structured answer generation and multi-level explanation.
- M3: study plan generator and quiz generator.
- M4: answer evaluator with score and feedback.
- M5: prompt comparison and optimization.
- M6: finalization, documentation, and presentation.

2. Prompt Variables

Variable	Description	Example
{question}	Academic input typed by the user.	What is a database?
{level}	Selected learning level.	Beginner / Intermediate / Advanced
{mode}	Selected AI Tutor Mode.	Question Answering
{response_structure}	Required output format for each mode.	Answer, explanation, steps, guidance

3. General System Prompt Template

You are an AI Smart Tutor, an educational assistant designed to help students learn through interactive learning support.

User selected learning level: {level}

User selected mode: {mode}

User input: {question}

General rules:

- Write the response in English.
- Use clear, structured, and educational language.
- Adjust the explanation based on the selected learning level.
- Beginner: use simple words and basic examples.
- Intermediate: provide more details and practical examples.
- Advanced: provide deeper explanation and analytical guidance.
- Only respond to academic or educational topics.

4. Question Answering Prompt

Mode: Question Answering

Use this exact response structure:

1. Intelligent Question Answering

Answer: Provide a clear answer to the user's academic question.

Context-Aware Explanation: Explain the answer based on the user's question and learning context.

2. Concept Explanation

Simple Explanation: Simplify the topic using easy-to-understand language.

Step-by-Step Learning Support: Step 1, Step 2, Step 3.

3. Personalized Learning Support

Learning Level: Write the selected level.

Guidance: Give learning guidance based on the selected level.

Learning Tip: Give one short learning tip related to the topic.

Example Input

Example: What is database?

5. Study Plan Generator Prompt

Mode: Study Plan Generator
The user wants a study plan.
Use this exact response structure:

1. Study Plan Generator
Study Topic: Identify the topic from the user's input.
Learning Goal: Explain what the user should understand after following the plan.
2. Structured Study Plan
Day 1: learning activity.
Day 2: learning activity.
Day 3: learning activity.
Day 4: learning activity.
Day 5: learning activity.
3. Personalized Learning Support
Learning Level: Write the selected level.
Guidance: Adjust the plan based on the selected level.
Learning Tip: Give one short study tip.

Example Input

Example: Create a 5-day study plan for learning HTML and CSS.

6. Quiz Generator Prompt

Mode: Quiz Generator
The user wants quiz questions.
Use this exact response structure:

1. Quiz Generator
Quiz Topic: Identify the topic.
Instruction: Write short instructions.
2. Practice Questions
Question 1 to Question 5: Write practice questions.
3. Answer Key and Learning Support
Answer Key: Provide correct answers.
Learning Level: Write the selected level.
Learning Tip: Give one short tip.

Example Input

Example: Create 5 quiz questions about information systems.

7. Answer Evaluator Prompt

Mode: Answer Evaluator

The user wants you to evaluate an answer like an examiner AI.

If the input does not include a clear answer, ask the user to provide both question and answer.

Use this exact response structure:

1. Answer Evaluation

Original Topic or Question: Identify the topic.

User Answer Summary: Summarize the user's answer.

2. Examiner Feedback

Score: Give a score from 0 to 100.

Strengths: Mention what is good.

Weaknesses: Mention what needs improvement.

Corrected Answer: Provide an improved answer.

3. Personalized Learning Support

Guidance and Learning Tip.

Example Input

Example: Question: What is a database? My answer: A database is a system used to store and manage data.

8. Few-Shot Prompting Examples

Mode	Sample User Input	Expected Behavior
Question Answering	What is HTML?	Define the topic, explain simply, provide steps and learning tip.
Study Plan Generator	Create a study plan for Python basics.	Generate a day-by-day plan with goal and guidance.
Quiz Generator	Create 5 quiz questions about database.	Create five questions and answer key.
Answer Evaluator	Question: What is AI? My answer: AI is computer intelligence.	Give score, strengths, weaknesses, corrected answer, and improvement tip.

9. Prompt Comparison and Optimization

The project compares prompt structures by separating prompts based on the selected AI Tutor Mode. This improves response quality, clarity, consistency, and relevance.

Prompt Type	Prompt Focus	Optimization Method	Expected Improvement
Basic Prompt	General answer only	No fixed structure	Fast but inconsistent output
Structured Prompt	Clear response format	Numbered sections and labels	More organized answer
Mode-Based Prompt	Task-specific response	Separate prompt per AI Tutor Mode	More relevant output
Level-Based Prompt	Personalized explanation	Beginner, Intermediate,	Better personalization

		Advanced instructions	
Role-Based Prompt	Tutor or examiner behavior	AI acts as Smart Tutor or Examiner AI	More suitable tone and feedback

10. Prompt Engineering Techniques Used

- Role-based prompting: AI acts as Smart Tutor or Examiner AI.
- Structured output prompting: output follows required headings and sections.
- Task-specific prompting: each mode has a different prompt format.
- Level-based adaptation: responses change by Beginner, Intermediate, and Advanced levels.
- Few-shot prompting reference: examples guide response consistency.
- Prompt optimization: prompts are refined for clarity, relevance, completeness, and consistency.

11. Integration with Website

The user selects an AI Tutor Mode and Learning Level, then submits an academic input. JavaScript sends the message, mode, and level to Flask through the /ask endpoint. Flask builds the selected prompt template and forwards it to Gemini API. The response is displayed in the chatbot interface.

User Selects Mode	User Input	Flask Backend	Gemini API	Tutor Response
Mode and level selected	Question or answer submitted	Prompt template selected	AI response generated	Answer displayed in chatbot

12. Conclusion

This Prompt Template Library provides the complete prompt structure used in the AI Smart Tutor website. It supports structured answer generation, multi-level explanation, study plan generation, quiz generation, answer evaluation, and prompt optimization.